Ahmed Saber Sakna

PhD Student, Johns Hopkins University — Project Controls, Water Resources & Remote Sensing
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linkedin.com/in/ahmad-sakna Professional Profile

Professional Summary

PhD student in Environmental Health and Engineering at **Johns Hopkins University** (satellite-based hydrology and terrestrial water storage) and project management professional specializing in sustainable water resources. Proven track record driving high-impact international programs within the **UNESCO, FAO, and Wageningen University (WUR)** frameworks, including co-developing climate adaptation initiatives awarded **€1,000,000 in AECID funding**. Expert in **Project Controls, Cost Estimation, and Spatial Data Analytics** utilizing **Oracle Primavera P6, SAP, PowerBI, Google Earth Engine, and WaPOR GIS tools**. Adept at navigating multi-ministry policies, technical assessments, and quantitative risk management to bridge rigorous science with large-scale operational impact.

Education

Johns Hopkins University, Whiting School of Engineering **Baltimore, MD, USA**
Ph.D. in Environmental Health and Engineering – Geography and Environmental Engineering Track *Aug. 2026 – Present*

- **Advisor:** Prof. Venkataraman Lakshmi. **Focus:** remote sensing, satellite-based hydrology, and terrestrial water storage.
- **Research:** (1) how soil-moisture initial-state quality limits 2–4 week **flash-drought forecast** skill; (2) detection limits of **radar–thermal fusion** for sensing irrigation water salinity in California's San Joaquin Valley.

IHE Delft Institute for Water Education **Delft, The Netherlands**
Master of Science in Water and Sustainable Development – Engineering Sciences *Oct. 2024*

- **Thesis:** *"Assessing the Impact of Climate Change on Water Consumption Using Remote Sensing Data and Machine Learning"* – Developed predictive ML models and high-level data reporting frameworks applicable to resource optimization.

Egyptian Russian University **Cairo, Egypt**
Bachelor of Science in Construction Engineering – **Ranked 1st in Cohort, GPA: 3.93/4.00** *July 2018*

Experience

Egyptian-Dutch Cooperation (FAO & Wageningen University) **Cairo, Egypt**
Project Consultant — Water Productivity (WaPOR) Project *Nov. 2026 – Present*

- Assess satellite-derived irrigation performance across pilot sites in the Nile Delta (Kafr El Sheikh, Shereshra, Hala, Baha, Sadat) for the FAO WaPOR Egypt project, as local consultant to Wageningen University & Research (WUR).
- **Technical report:** Built a multi-site Python/Google Earth Engine pipeline with a Difference-in-Differences framework, and authored the master technical report (49+ figures) after a full data-integrity audit.
- **Literature review:** Authored the literature review for FAO WaPOR deliverables D1.1/D1.2, with original analysis of validation circularity and the limits of observing irrigation efficiency from space.
- **Learning sessions:** Delivered learning sessions on applying WaPOR data to irrigation performance assessment for project partners and Egyptian water-sector stakeholders.
- Developed national guidelines and policy recommendations for operational use and scaling of WaPOR across Egypt.

United Nations Educational, Scientific and Cultural Organization (UNESCO) **Cairo, Egypt**
Individual Consultant — Regional Office for Egypt & Sudan *Oct. 2024 – Present*

- **Co-developed two climate adaptation projects awarded €1,000,000 from AECID**, leading the financial planning, budget breakdown, and milestone scheduling for green economy and coastal resilience initiatives.
- Developed automated **PowerBI dashboards** for executive reporting to track project KPIs, funding disbursement, and resource allocation across diverse desert and coastal biomes.
- Secured an additional $50,000+ in donor funding for a regional Water-Energy-Food Nexus Competition; managed the cost control and logistics for capacity-building events reaching 500+ stakeholders.
- Coordinated a 12-country Expert Group (30+ specialists), delivering geospatial and financial progress reports for the UNESCO MAB Meeting (China) to influence regional conservation policies.
- Facilitated coordination across 5 national ministries to align procurement policies and execute interdisciplinary programs within strict international donor timelines.

Hassan Allam Construction **ElDabaa, Egypt**

Project Engineer — ElDabaa Nuclear Power Plant *June 2022 – Oct. 2023*

- **Planning & Scheduling:** Utilized **Oracle Primavera P6** to develop and update baseline schedules for water supply and firefighting networks for 174+ buildings within the $1 billion project.
- **Cost Control & Reporting:** Spearheaded risk and schedule management initiatives, achieving a **15% reduction in cost overruns**; generated weekly and monthly progress reports (MPR) using **PowerBI** for multinational stakeholders.
- **Resource Management:** Managed material procurement and inventory tracking through **SAP**, ensuring seamless supply chain operations for Egypt's largest concrete batch plant (240 m^3/hour).
- **Stakeholder & Quality Control:** Coordinated with 5 multinational stakeholders to monitor environmental risk and project progress, ensuring 100% compliance with nuclear QA/QC and safety standards.

Egyptian Russian University **Cairo, Egypt**

Teaching Assistant — Water Resource Management & Project Management *Sep. 2018 – June 2022*

- Designed and delivered curriculum for 120+ students, emphasizing **Project Management** fundamentals including CPM scheduling, cost estimation, and Earned Value Management (EVM).
- Developed new course material integrating **MS Project** and GIS tools, modernizing instruction to align with international construction industry practices.

Technical Skills

Project Controls & Planning: **Oracle Primavera P6**, **SAP (ERP)**, **PowerBI Reporting**, Cost Estimation, Risk Management, EVM, MS Project
Engineering & Data: **SAP2000**, ETABS, Python (Pandas, Scikit-learn), SQL, Advanced Excel (Dashboards), AutoCAD, LaTeX
GIS & Remote Sensing: **Google Earth Engine**, QGIS, ArcGIS, WaPOR/AgERA5 analysis, SMAP, Sentinel-1, NISAR, GRACE-FO, CMIP6, Spatial Analysis
Certifications: **PMP (Expected 2026)**, Google Project Management Professional, Risk Management Practical Workshop

Research Work & Publications

- Sakna, A.S., et al. (2026). Assessing the impact of climate change on agricultural water consumption using earth observations and machine learning: a case study of Zankalon, Nile Delta, Egypt. *Journal of Hydrology* (under review).
- Sakna, A.S., and Lakshmi, V. (2026). Initial soil-moisture state quality and flash-drought forecast predictability. *AGU Fall Meeting 2026* (abstract submitted, session H074).
- Sakna, A.S., and Lakshmi, V. (2026). Satellite detection limits for irrigation water quality via radar–thermal fusion. *AGU Fall Meeting 2026* (abstract submitted, session H106).
- Abdelal, Q., et al. (2026). Reduced Frequency of Soil Moisture Measurements for **Cost-effective** Irrigation Management. *Journal of Agrometeorology.*
- Galishnikova, V.V., et al. (2021). Self-healing bacterial mortar with calcium lactate and improved properties. *Magazine of Civil Engineering.*

Languages

Arabic: Mother tongue **English**: Fluent (IELTS Band 8.0) **Russian**: Intermediate **French**: Intermediate